Section 4.3

(1) $X \sim \text{Poisson}(4)$

(a) $P(X=1) = \dfrac{e^{-4}\,4^1}{1!}$

$\boxed{P(X=1) = 0.0733}$

(b) $P(X=0) = \dfrac{e^{-4}\,4^0}{0!}$

$\boxed{P(X=0) = 0.0183}$

(c) $P(X<2) = P(X=0) + P(X=1)$

$= \dfrac{e^{-4}\,4^0}{0!} + \dfrac{e^{-4}\,4^1}{1!}$

$\boxed{P(X<2) = 0.0916}$

(d) $P(X>1) = 1 - P(X \leq 1)$

$= 1 - [P(X=0) + P(X=1)]$

$= 1 - 0.0916 = 0.9084$

(e) $\mu_X = \lambda = 4$

(f) $\sigma_X^2 = \lambda \Rightarrow \sigma_X = \sqrt{\lambda} = \sqrt{4} = 2$

$\boxed{\sigma_X = 2}$

(3)

$X \sim \text{Poisson}(5)$

(a) $P(x=6)$

$$= \frac{e^{-5}\,5^6}{6!}$$

$$= 0.1462$$

(b) $P(x \leq 2) = P(x=0) + P(x=1) + P(x=2)$

$$= \frac{e^{-5}\,5^0}{0!} + \frac{e^{-5}\,5^1}{1} + \frac{e^{-5}\,5^2}{2!}$$

$$= 0.1247$$

(c) $P(5 < x < 8)$

$$= P(x=6) + P(x=7)$$

$$= \frac{e^{-5}\,5^6}{6!} + \frac{e^{-5}\,5^7}{7!}$$

$$= 0.2507$$

(d) $\mu_x = \lambda = 5$

$$\boxed{\mu_x = 5}$$

(e) $\sigma_x = \sqrt{\sigma_x^2} = \sqrt{\lambda} = \sqrt{5}$

$$\boxed{\sigma_x = \sqrt{5} = 2.2361}$$

(3)

(a) $\lambda = 4$

$X \sim \text{poisson}(4)$

(a) $P(X=3) = \frac{e^{-4}\, 4^3}{3!}$

$= 0.1954$

(b)

$\lambda = 3 \cdot 4 = 12$

$X \sim \text{poisson}(12)$

$P(X=8) = \frac{e^{-12}\, 12^8}{8!}$

$= 0.0655$

(c) $\lambda = 3 \cdot 2 = 6$

$X \sim \text{poisson}(6)$

$P(X>3) = 1 - P(X \leq 3) = 1 - P(X=0) - P(X=1) - P(X=2) - P(X=3)$

$= 1 - \frac{e^{-6}\, 6^0}{0!} - \frac{e^{-6}\, 6^1}{1} - \frac{e^{-6}\, 6^2}{2 \cdot 1} - \frac{e^{-6}\, 6^3}{3 \cdot 2 \cdot 1}$

$P(X>3) = 0.8488$

(13) $X \sim \text{poisson}(20)$

(a) $P(x=15) = \dfrac{e^{-20}\,20^{15}}{15!}$

$= 0.0516$

(b) $x \sim Bin(15, 0.6)$

$P(x=10) = \binom{15}{10} (.6)^{10} (1-.6)^{5}$

$= 0.1859$

(c) $x \sim Bin(N, 0.6)$

(d) $P(x=15)$ and $P(x=10)$

$= 0.0516\,(0.1859) = 0.00960$

(15) $x \sim \text{poisson}(0.45 * 3)$

(a) $P(x=0) = \dfrac{e^{-1.35}\,1.35^{0}}{0!}$

$= 0.2592$

(b) $P(x=0) = e^{0.45n}$ $= 0.5$

$n(0.45) = \ln(.5) \Rightarrow n = \dfrac{\ln(.5)}{.45}$

$n = 1.54\ m$

(b)

(a) $\lambda = \frac{100}{60} = \frac{5}{3}$

$X \sim \text{poisson}\ (5/3)$

$P(x=0) = \frac{e^{-5/3}\ (5/3)^0}{0!} = 0.1889$

(b) 200

$P(x=5) = \frac{e^{-10/3}\ (10/3)^5}{5!}$ $\quad \lambda = \frac{200}{60} = \frac{10}{3}$

$= 0.1223$

(c) $P(x=0) = 0.01 = \frac{e^{-n/60}\ (n/60)^0}{0!}$

$e^{-n/60} = 0.01 \Rightarrow n/60 = \ln(0.01)$

$-n = 60\ (\ln(0.01))$

$n = 276.3$

$\therefore$ Atleast 277 raisins needed.

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(a) $\mu_x = \frac{14 + 11}{2} = 12.5$

(b) $\mu_y = \frac{6 + 8}{2} = 7$

(c) $\sigma_x = \sqrt{\frac{\lambda_x}{t}} = \sqrt{\frac{12.5}{2}} = 2.5$

$\sigma_x = 2.5$

(d) $\sigma_y = \sqrt{\frac{\lambda_y}{t}} = \sqrt{\frac{7}{2}} = 1.87 \simeq 1.9$

(e) $\mu_{x-y} = \mu_x - \mu_y = 12.5 - 7 = 5.5$

$\sigma_{x-y} = \sqrt{\sigma_x^2 + \sigma_y^2} = \sqrt{2.5^2 + 1.9^2} = \pm 3.1$

$\mu_{x-y} = 5.5$ $\sigma_{x-y} = \pm 3.1$